

PIZZA SALES

ANALYSIS

PROJECT



Hello, my name is **Vipul Sharma**, I have a strong interest in data analysis and business intelligence. This project reflects my skills in **SQL** and **data analysis**, where I analyzed pizza sales data to discover valuable business insights and sales trends.



Salford & Co.

ABOUT OUR PROJECT

About Us

This project is based on Pizza Sales Analysis using SQL. The main objective of the project is to analyze pizza sales data and gain meaningful business insights by using different SQL queries. The project helps in understanding customer ordering behavior, sales performance, revenue generation, and product demand.

In this project, various levels of SQL queries such as Basic, Intermediate, and Advanced queries are used to explore the dataset effectively. The analysis includes finding the total number of orders, total revenue generated, highest-priced pizza, most common pizza size, and top-selling pizza types. It also focuses on category-wise sales analysis, hourly order distribution, average daily orders, and revenue-based performance.

Advanced analysis techniques are used to calculate the percentage contribution of each pizza type to total revenue, cumulative revenue over time, and the top-performing pizzas in each category. These insights help businesses understand customer preferences, improve decision-making, manage inventory, and increase profitability.

Overall, this project demonstrates practical knowledge of SQL, data analysis, joins, aggregation functions, window functions, grouping, and business intelligence concepts through a real-world pizza sales dataset.



RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

Result Grid 	
	total_orders
▶	21350



CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT  
    ROUND(SUM(orders_details.quantity * pizzas.price),  
          2) AS Total_Pizza_Sales  
FROM  
    orders_details  
    JOIN  
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
```

Result Grid			
	Total_Pizza_Sales		
▶	817860.05		



IDENTIFY THE HIGHEST-PRICED PIZZA.

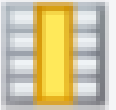
```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid			Filter Rows
	name	price	
▶	The Greek Pizza	35.95	



IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
    pizzas.size, COUNT(orders_details.order_details_id)
FROM
    pizzas
    JOIN
        orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizzas.size
ORDER BY pizzas.size
LIMIT 1;
```

Result Grid   Filter Rows:		
	size	COUNT(orders_details.order_details_id)
▶	L	18526



LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid   Filter Rows:		
	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371



JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
SELECT
    SUM(orders_details.quantity) AS quantity,
    pizza_types.category
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid   Filter Rows		
	quantity	category
▶	14888	Classic
	11987	Supreme
	11649	Veggie
	11050	Chicken



DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    HOUR(order_time) AS hours, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

Result Grid			Filter Rows
	quantity	category	
▶	14888	Classic	
	11987	Supreme	
	11649	Veggie	
	11050	Chicken	



JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category  
ORDER BY COUNT(name) DESC;
```

Result Grid			Filter Rows:
	category	COUNT(name)	
▶	Supreme	9	
	Veggie	9	
	Classic	8	
	Chicken	6	



GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```
SELECT
    AVG(quantity)
FROM
    (SELECT
        orders.order_date, SUM(orders_details.quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid		Filter Rows:	
	AVG(quantity)		
▶	138.4749		



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT
    pizza_types.name,
    SUM(orders_details.quantity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas
    ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN orders_details
    ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid			Filter Rows:
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	



CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
SELECT
  pizza_types.category,
  ROUND(
    (SUM(orders_details.quantity * pizzas.price) /
    (SELECT
      SUM(orders_details.quantity * pizzas.price)
      FROM orders_details
      JOIN pizzas ON pizzas.pizza_id = orders_details.pizza_id)
    ) * 100, 2
  ) AS revenue_percentage
FROM
  pizza_types
JOIN
  pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN
  orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY
  pizza_types.category
ORDER BY
  revenue_percentage DESC;
```

Result Grid



Filter Rows:

	category	revenue_percentage
▶	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68



ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
SELECT
    order_date,
    SUM(revenue) OVER(ORDER BY order_date) AS cum_revenue
) FROM (
    SELECT
        orders.order_date,
        SUM(orders_details.quantity * pizzas.price) AS revenue
    FROM orders_details
    JOIN pizzas ON orders_details.pizza_id = pizzas.pizza_id
    JOIN orders ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date
) AS sales;
```

Result Grid			Filter Rows:
	order_date	cum_revenue	
	2015-08-03	492610.60000000003	
	2015-08-04	494700.750000000035	
	2015-08-05	496795.60000000003	
	2015-08-06	498894.85000000003	
	2015-08-07	501521.250000000035	
	2015-08-08	504237.65000000004	
	2015-08-09	506240.30000000004	
	2015-08-10	508379.75000000004	
	2015-08-11	510669.75000000004	
	2015-08-12	513035.50000000004	
	2015-08-13	515109.650000000043	
	2015-08-14	518126.25000000004	
	2015-08-15	520378.60000000004	
	2015-08-16	522517.90000000004	
	2015-08-17	525143.90000000004	
	2015-08-18	527245.10000000003	
	2015-08-19	529578.05000000003	
	2015-08-20	531485.75000000002	
	2015-08-21	534087.15000000003	
	2015-08-22	536493.15000000003	
	2015-08-23	538194.75000000002	
	2015-08-24	539891.80000000003	
Result 1			×



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
SELECT category,name, revenue FROM (  
  SELECT  
    category, name, revenue,  
    RANK() OVER(PARTITION BY category ORDER BY revenue DESC) AS rn  
  FROM (  
    SELECT  
      pizza_types.category, pizza_types.name,  
      SUM(orders_details.quantity * pizzas.price) AS revenue  
    FROM pizza_types  
    JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id  
    JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id  
    GROUP BY pizza_types.category, pizza_types.name) AS a) AS b  
WHERE rn <= 3;
```

	category	name	revenue
▶	Chicken	The Thai Chicken Pizza	43434.25
	Chicken	The Barbecue Chicken Pizza	42768
	Chicken	The California Chicken Pizza	41409.5
	Classic	The Classic Deluxe Pizza	38180.5
	Classic	The Hawaiian Pizza	32273.25
	Classic	The Pepperoni Pizza	30161.75
	Supreme	The Spicy Italian Pizza	34831.25
	Supreme	The Italian Supreme Pizza	33476.75
	Supreme	The Sicilian Pizza	30940.5
	Veggie	The Four Cheese Pizza	32265.700000000065
	Veggie	The Mexicana Pizza	26780.75
	Veggie	The Five Cheese Pizza	26066.5

PROJECT HIGHLIGHTS & KEY INSIGHTS

What I Learned

- Learned how to analyze real-world sales data using SQL.
- Improved skills in joins, aggregate functions, subqueries, and window functions.
- Understood customer purchasing behavior and sales trends.
- Learned how to convert raw data into meaningful business insights.
- Gained practical experience in data analysis and business decision-making.

Analysis Performed

- Total orders and total revenue analysis.
- Identified highest-priced and most ordered pizzas.
- Analyzed pizza size and category-wise sales distribution.
- Studied hourly and daily ordering patterns.
- Found top-performing pizzas based on quantity and revenue.
- Calculated revenue contribution and cumulative sales trends.

Suggestions to Increase Sales

- Promote top-selling pizzas through special offers and combos.
- Provide discounts during low-sales hours to attract customers.
- Focus marketing on less-selling pizza categories.
- Introduce loyalty programs and festive offers for repeat customers.
- Improve inventory management based on demand trends.
- Use customer preferences data to create targeted promotions



THANK YOU



Thank you for viewing this project. Your valuable feedback and suggestions are highly appreciated as they will help me improve my skills and future projects.

